



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Fall 2024-2025

Section: [L]

Group No: 08

Project Report On

Project Name [Pharmacy Management System]

Supervised By

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Submitted By:

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Obtained Marks for CO2 and CO3 (Description given in the following page)						
Assessment Criteria		Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)		Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment				Organization of the application		
Validation				Representation and Integration of Database		
Verification				Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are

	implementation of verification processes, and data flow is not considered.	incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	attention to ensuring data integrity and appropriate data flow within the system.	comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way

		identified as coherent.			according to the coherent usage .
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CHAPTER 01: INTRODUCTION

The Pharmacy Management System is a computer-based solution to the problems associated with the manual operation of a pharmacy. It efficiently manages medicines, transactions, and customer relationships. It is a platform where admin, cashiers, and customers can efficiently manage sales and orders records. As the demand for pharmaceutical products is increasing day by day, it becomes tedious to handle stock levels, transactions, and customer records manually. A well-designed and structured Pharmacy Management System is meant to bring ease to its operation, thereby improving efficiency, cashiers and customer experience.

CHAPTER 02: User Stories

User Story: Admin		
Use case name	Actor	User story
Sign in	Admin	As an admin, I want to Sign In using username and Password word etc. So that I can securely access the system.
Log In	Admin	I want to login to the System so that I can manage the Pharmacy operation.
Dashboard	Admin	I want to view key metrics like total cashier, total order, Today's revenue and total revenue. So that I can monitor the systems' performance.
Add user	Admin	I want to add, update, or delete. Cashier or other accounts so that I can manage system users.
Categories	Admin	I want to create, update and delete product categories, so that the inventory is well organized.
Add product	Admin	I want to add, update or delete product details so that I can keep the Inventory accurate.
Transactions	Admin	I want to view detailed all transaction records, so that I can track sales and revenue.
Logout	Admin	I want to log out securely, so that I can Protect my account.

User Stories: Cashier		
Use case Name	Actor	User story
Sign In	Cashier	As a cashier, I want to sign in with my username and Password. So that I can access my account Securely.
Log In	Cashier	I want to log in to the system if admin approve, so that I can Process customer orders.
Orders	Cashier	I want to Process customers order selecting items, quantity and calculating total price, cancel orders and print receipts for efficient Service.
Dashboard	Cashier	I want to view and overview of like total cashier, total order, Today's revenue and total revenue. So that I can monitor the system's performance.
Transactions	Cashier	I want to view all completed transactions for the day to ensure accuracy and accountability.

Log Out	Cashier	I want to log out securely, so that I can Protect my account information.
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User Stories: Customer		
Use case Name	Actor	User story
Sign In	Customer	As a customer, I want to sign in to my account so that I can track my purchase history.
Order	Customer	I want to view available products with categories and prices, add for an order. I want to choose product quantities and view Total cost to prepare for pay.
Pay	Customer	I want to check the total Price. Paid amount and remain balance to ensure accurate Transaction. Also Received Money receipt.
Log Out	Customer	I want to log out securely, so that I can protect my account information.

Chapter 3: Database

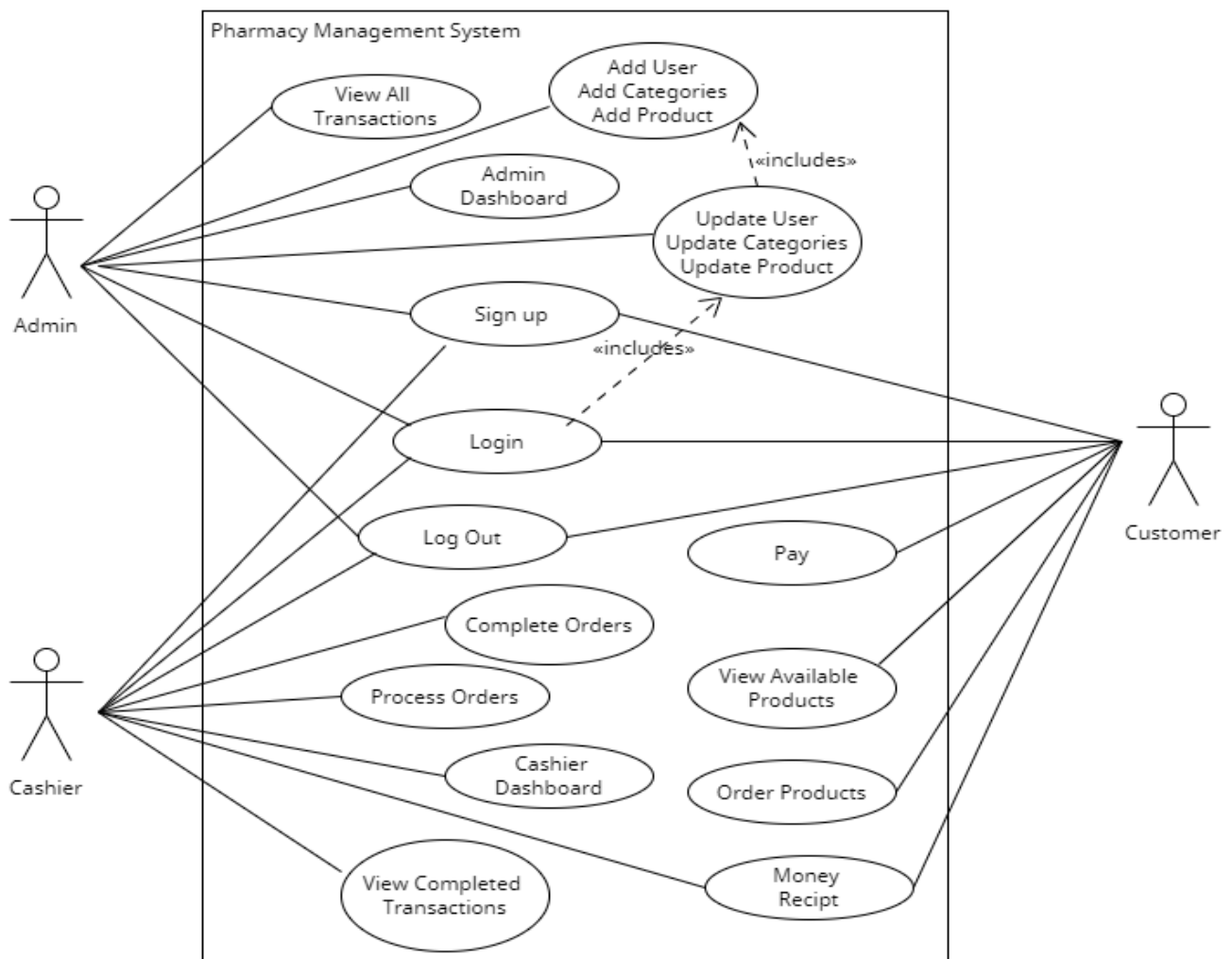


Fig: Use Case diagram

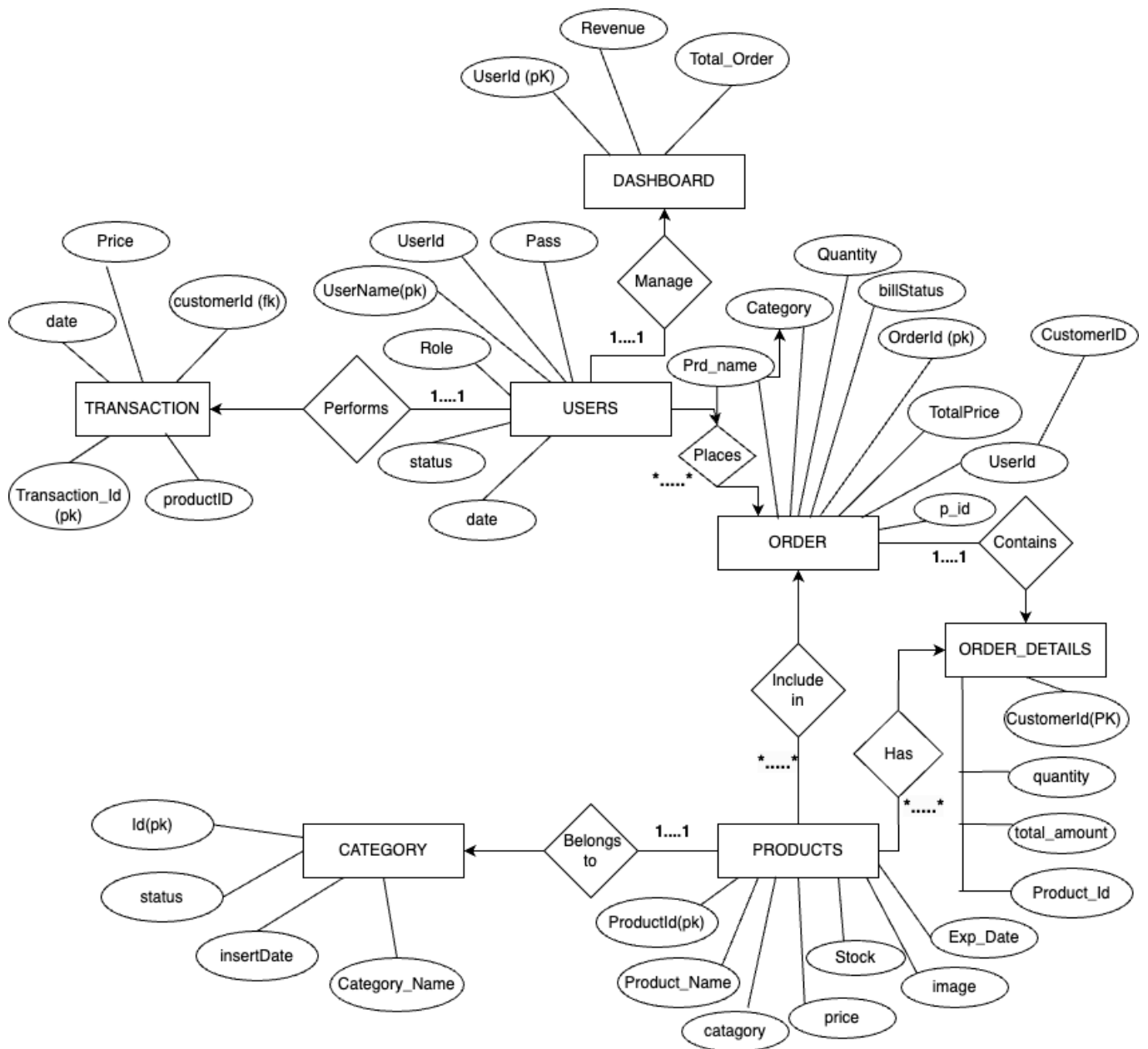


Fig: ER Diagram

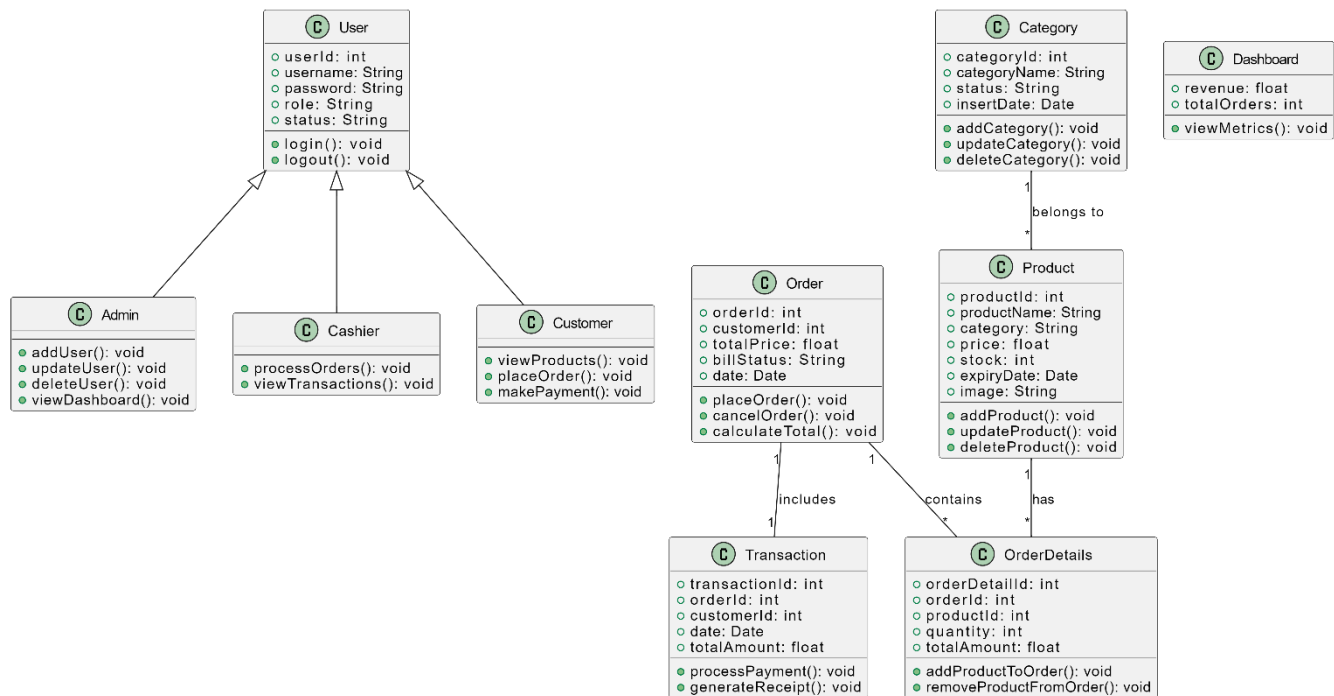


Fig: Class Diagram

User

Key	Name	Data Type	Length	Nullable
	Id	INTEGER	10	No
PK	userName	VARCHAR	50	No
	Password	VARCHAR	50	No
	Role	VARCHAR	50	No
	dateregister	Date	-	Yes

Category

Key	Name	Data Type	Length	Nullable
	Id	INTEGER	10	No
PK	category	VARCHAR	50	No
	Status	VARCHAR	50	No
	insertDate	Date	-	No

Product

Key	Name	Data Type	Length	Nullable
	id	INTEGER	10	No
PK	productId	VARCHAR	10	No
	productName	VARCHAR	50	No
	stocks	INTEGER	10	No
	price	INTEGER	10	No
FK	category	VARCHAR	50	No
	image	VARCHAR	255	No
	expairDate	Date	-	No

Order

Key	Name	Data Type	Length	Nullable
PK	Id	INTEGER	10	No
FK	customerID	INTEGER	10	No
FK	productName	VARCHAR	50	No
FK	productId	VARCHAR	10	No
FK	category	VARCHAR	50	No
	quantity	INTEGER	10	No
	price	INTEGER	10	No
	totalPrice	INTEGER	10	No
	status	VARCHAR	50	No
	orderdate	Date	-	No
	billstatus	VARCHAR	50	No

SQL queries

Create Database	CREATE DATABASE Pharmacy_Management_System;
Create Users Table	<pre>CREATE TABLE users id INTEGER(10) IDENTITY (1,1), username VARCHAR(50) NOT NULL PRIMARY KEY, Password VARCHAR(50) NOT NULL, Role VARCHAR(50) NOT NULL, dateRegister DATE NULL);</pre>
Insert Admin Credentials	<pre>INSERT INTO users (username, password, role, status, dateRegister) VALUES ('maruf', '1234', 'Admin', 'Active', '2025-01-20');</pre>
Create Categories Table	<pre>CREATE TABLE Category (Id INTEGER(10) NOT NULL IDENTITY(1,1), category VARCHAR(50) NOT NULL PRIMARY KEY, Status VARCHAR(50) NOT NULL, insertDate DATE NOT NULL);</pre>
Create Products Table	<pre>CREATE TABLE Products (id INTEGER(10) NOT NULL IDENTITY (1,1), productId VARCHAR(10) NOT NULL PRIMARY KEY, productName VARCHAR(50) NOT NULL, stocks INTEGER(10) NOT NULL, price INTEGER(10) NOT NULL, category VARCHAR(50) NOT NULL, image VARCHAR(255) NOT NULL, expairDate DATE NOT NULL);</pre>

Create Orders Table	<pre> CREATE TABLE Order (Id INTEGER (10) NOT NULL PRIMARY KEY IDENTITY (1,1), customerID INTEGER (10) NOT NULL, productName VARCHAR (50) NOT NULL, productID VARCHAR (10) NOT NULL, category VARCHAR (50) NOT NULL, quantity INTEGER (10) NOT NULL, price INTEGER (10) NOT NULL, totalPrice INTEGER (10) NOT NULL, status VARCHAR (50) NOT NULL, orderdate DATE NOT NULL, billstatus VARCHAR (50) NOT NULL, FOREIGN KEY (customerID) REFERENCES users (Id), FOREIGN KEY (productID) REFERENCES Products(productID), FOREIGN KEY (category) REFERENCES Category (category)); </pre>
Join Query for admin Transactions	<pre> SELECT u.username AS CustomerName, o.customerID, o.productID, o.price, o.totalprice, o.orderdate AS transactionDate FROM orders o INNER JOIN users u ON o.customerID = u.id WHERE o.billstutas = 'complete'; </pre>
For Cashier Transactions	<pre> SELECT * FROM orders WHERE billstutas = 'complete' AND orderdate = CAST(GETDATE () AS DATE); </pre>

Chapter 4: Screenshot of forms with figure names:



Fig: Loding Page

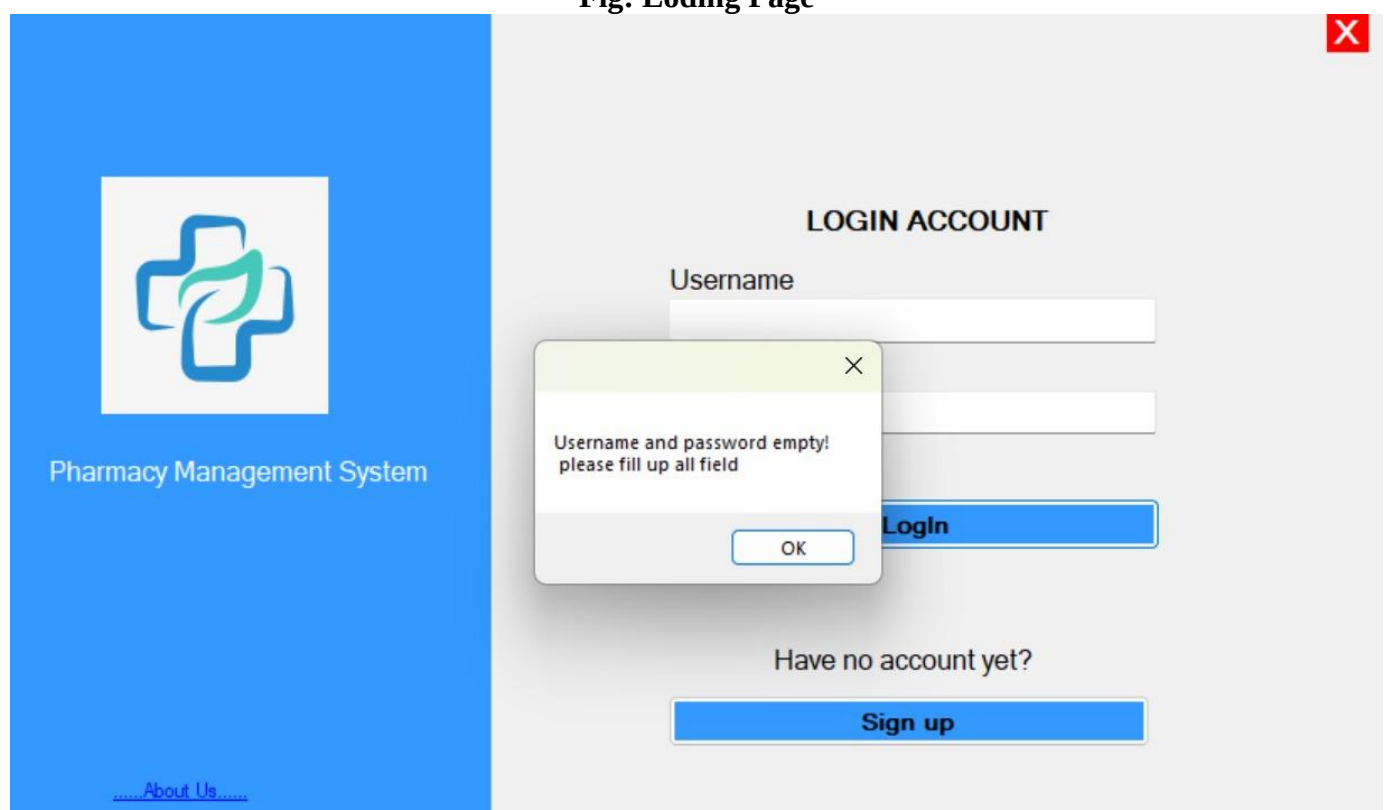


Fig: Login Page

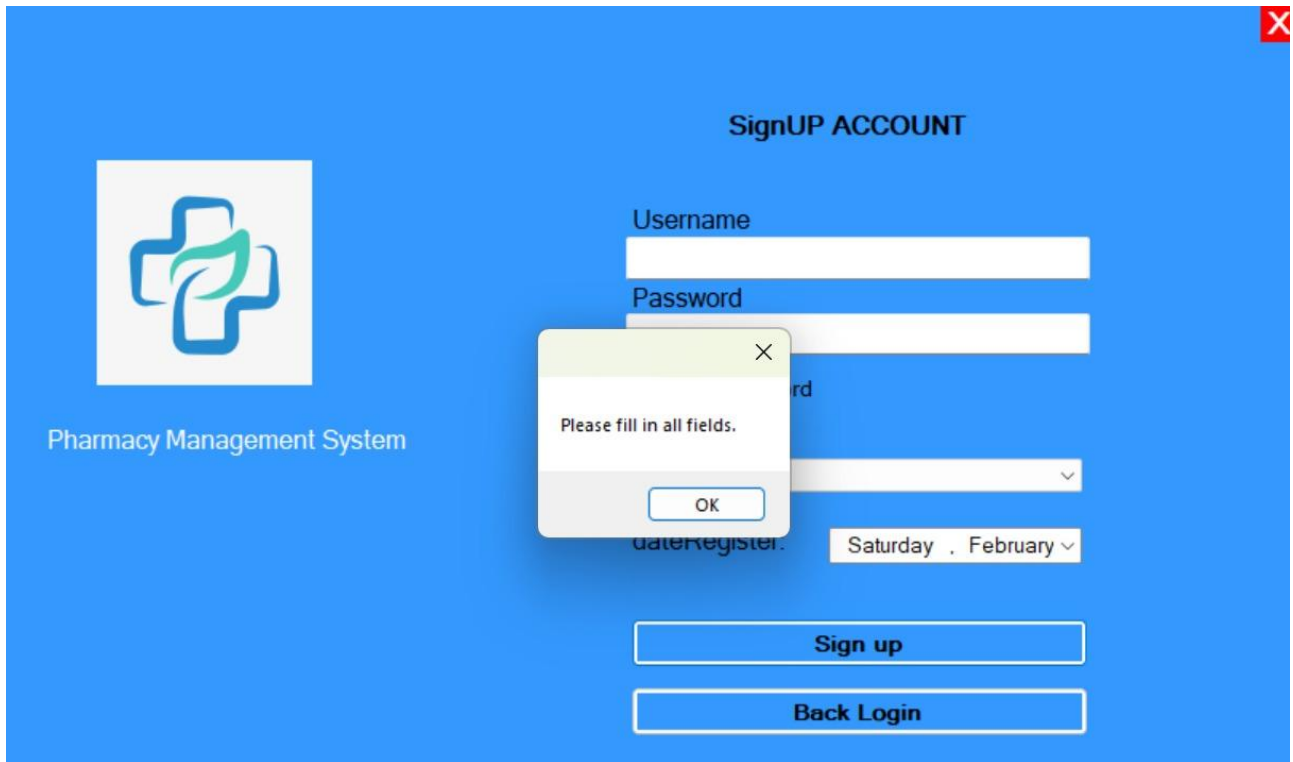


Fig: Sign up Page

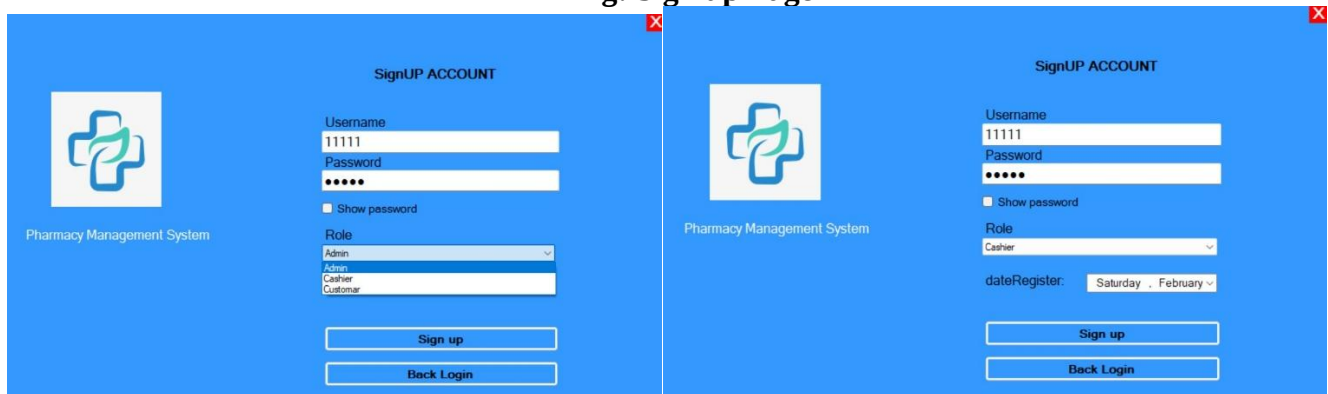


Fig: Sign up Dynamic Control

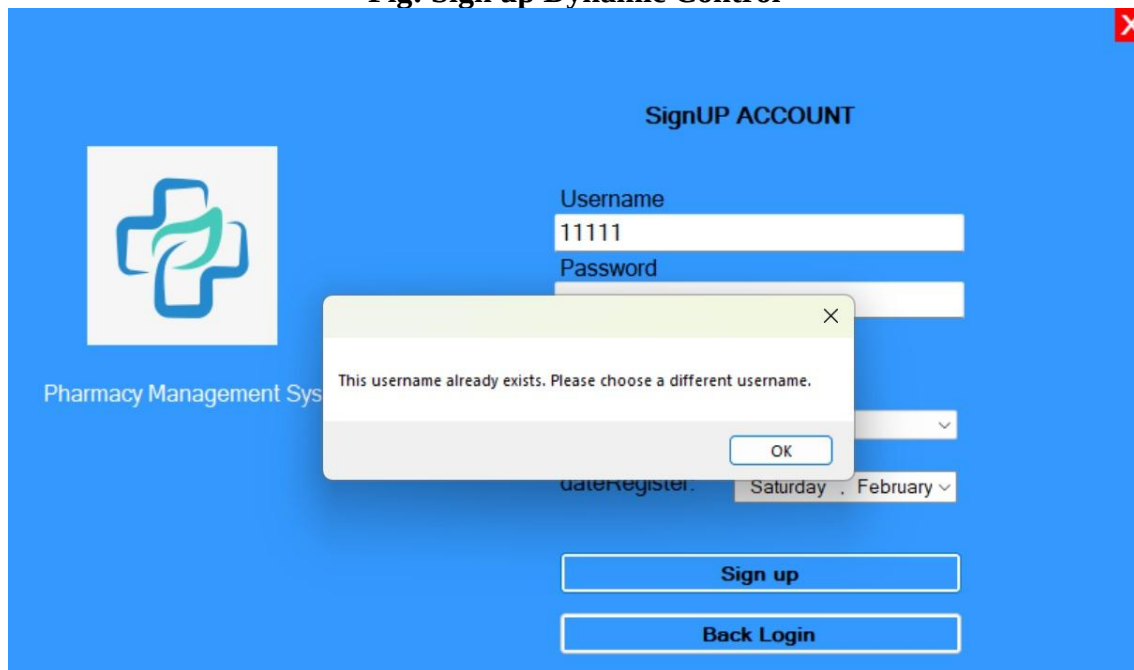


Fig: Sign up Page With Error

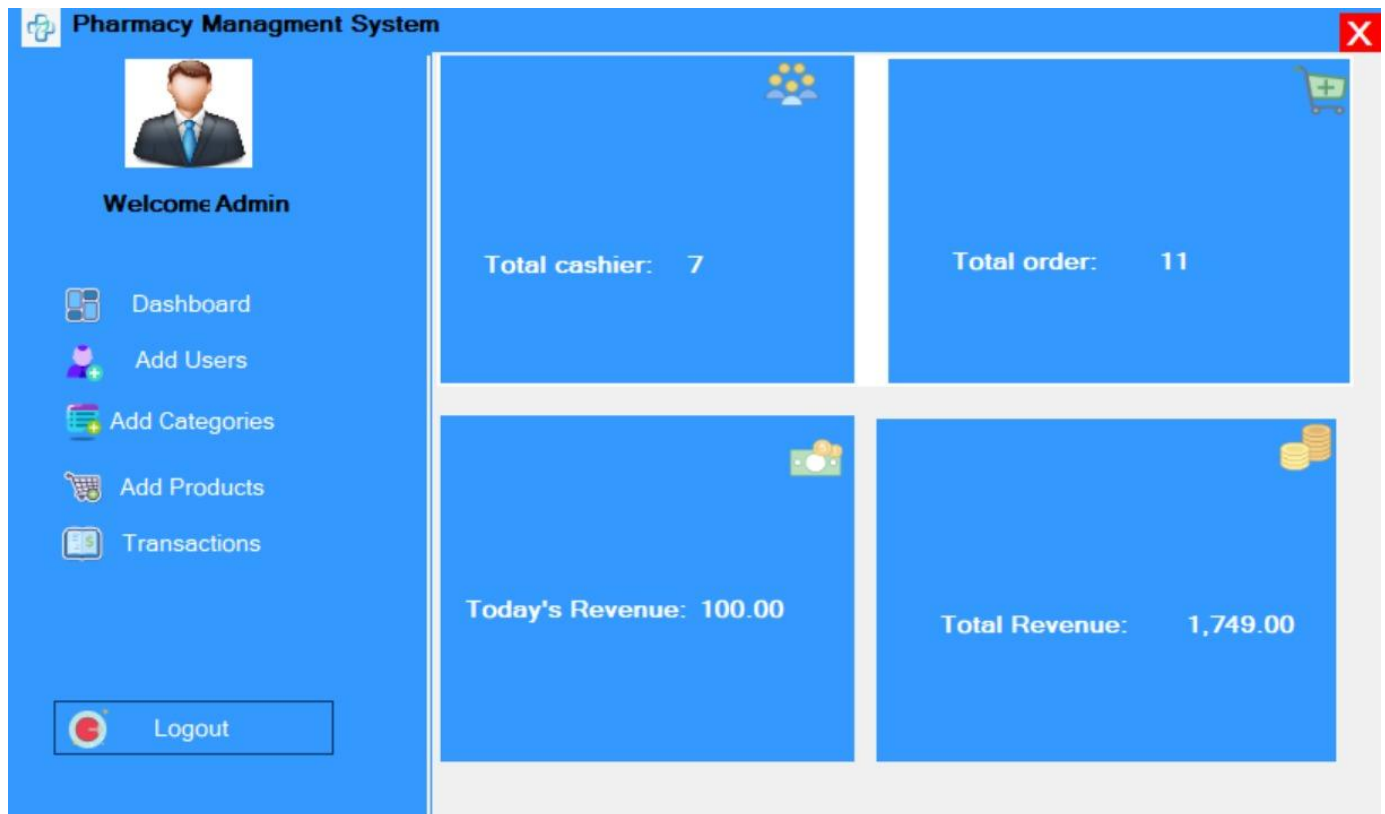


Fig: Admin Dashboard



Fig: Add Users


Pharmacy Management System



Welcome Admin

- Dashboard
- Add Users
- Add Categories
- Add Products
- Transactions

Logout




Category

Status

ADD

UPDATE

DELETE

CLEAR

Data

	category	status	insert
▶	Drops (For Eyes, ...	Active	2/5/2
	Inhalers & Nebuli...	Active	2/5/2
	Injections	Active	2/5/2
	Ointments & Crea...	Active	2/5/2
	Powders & Granu...	Active	2/5/2
	qwe	Active	2/6/2
	Suppositories	Active	2/5/2
	Syrups	Active	2/5/2
	Tablets	Active	2/5/2
*			

Fig: Add Category


Pharmacy Management System



Welcome Admin

- Dashboard
- Add Users
- Add Categories
- Add Products
- Transactions

Logout

product Data

	id	productID	productName	price	stock
▶	41	P001	Napa	10	990
	42	P002	Ace	12	0
	43	P003	Napa Syrup	25	59
	44	P004	Panadol	8	190

product Id

product Name

price(TK)

Stock

Catagory

Status

expair date

Saturday , February

ADD

UPDATE

DELETE

CLEAR

Import

Fig: Add Product



Fig: Admin Transactions

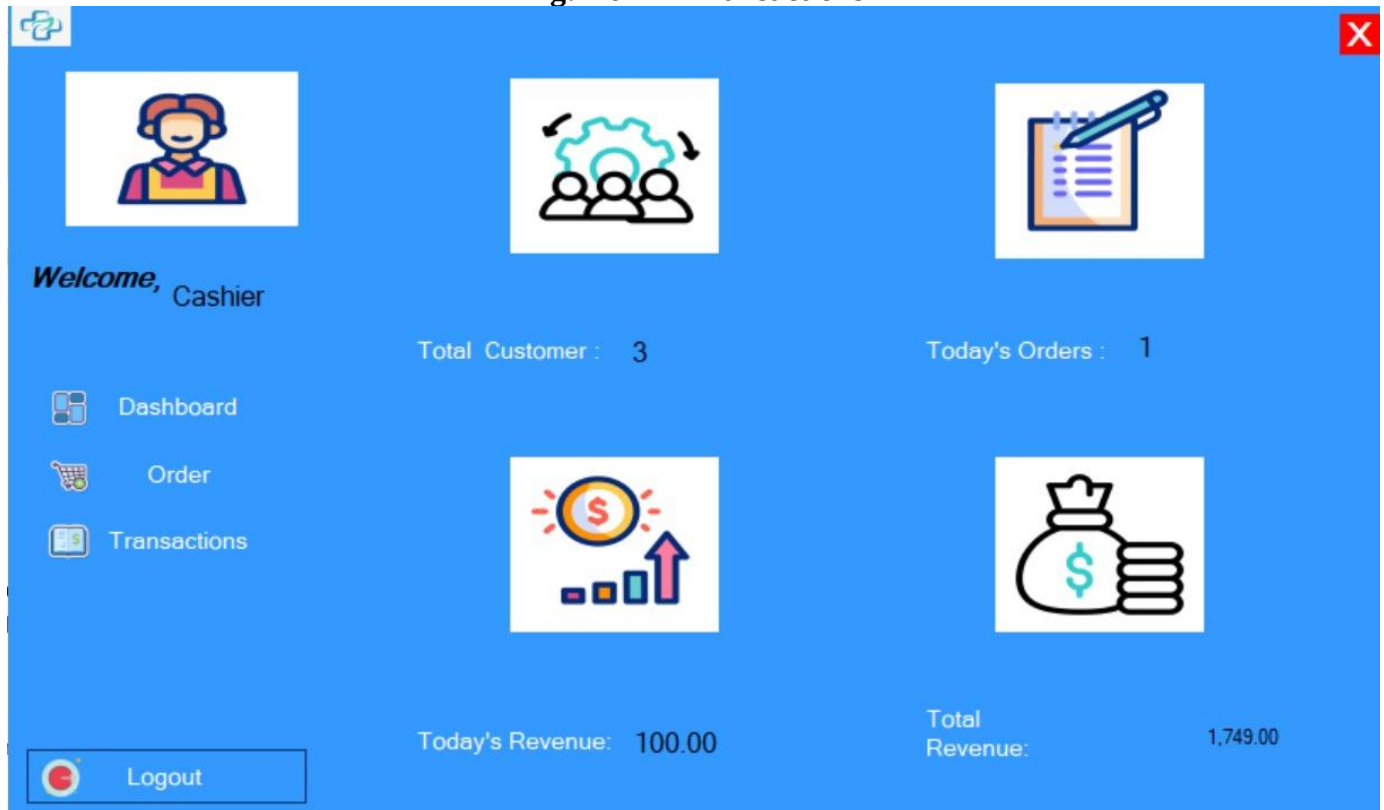
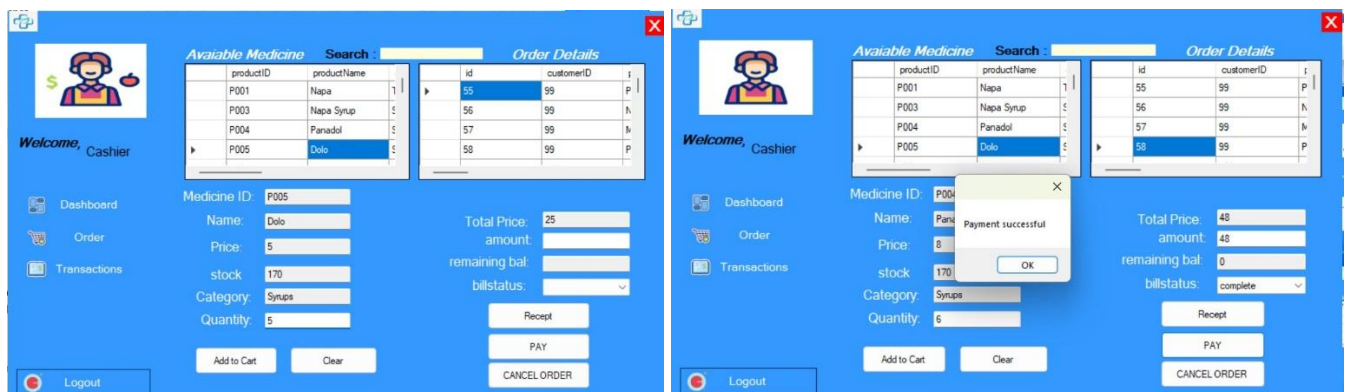


Fig: Cashier Dashboard



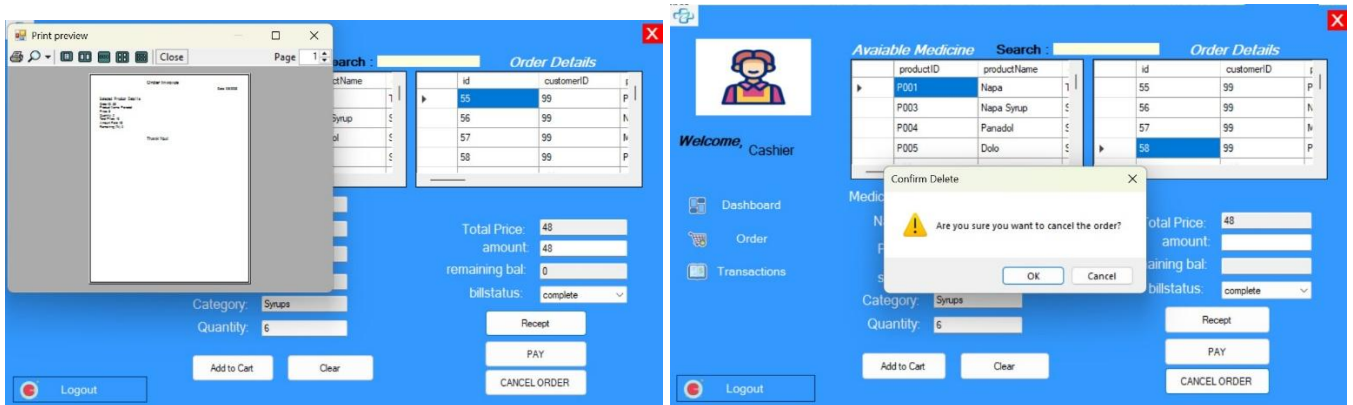


Fig: Cashier Orders

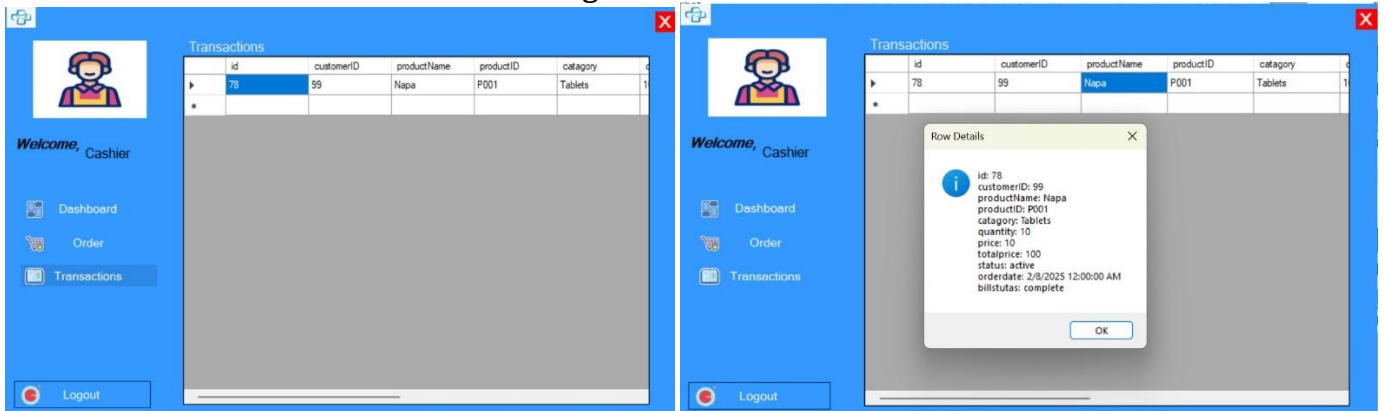


Fig: Cashier Transactions

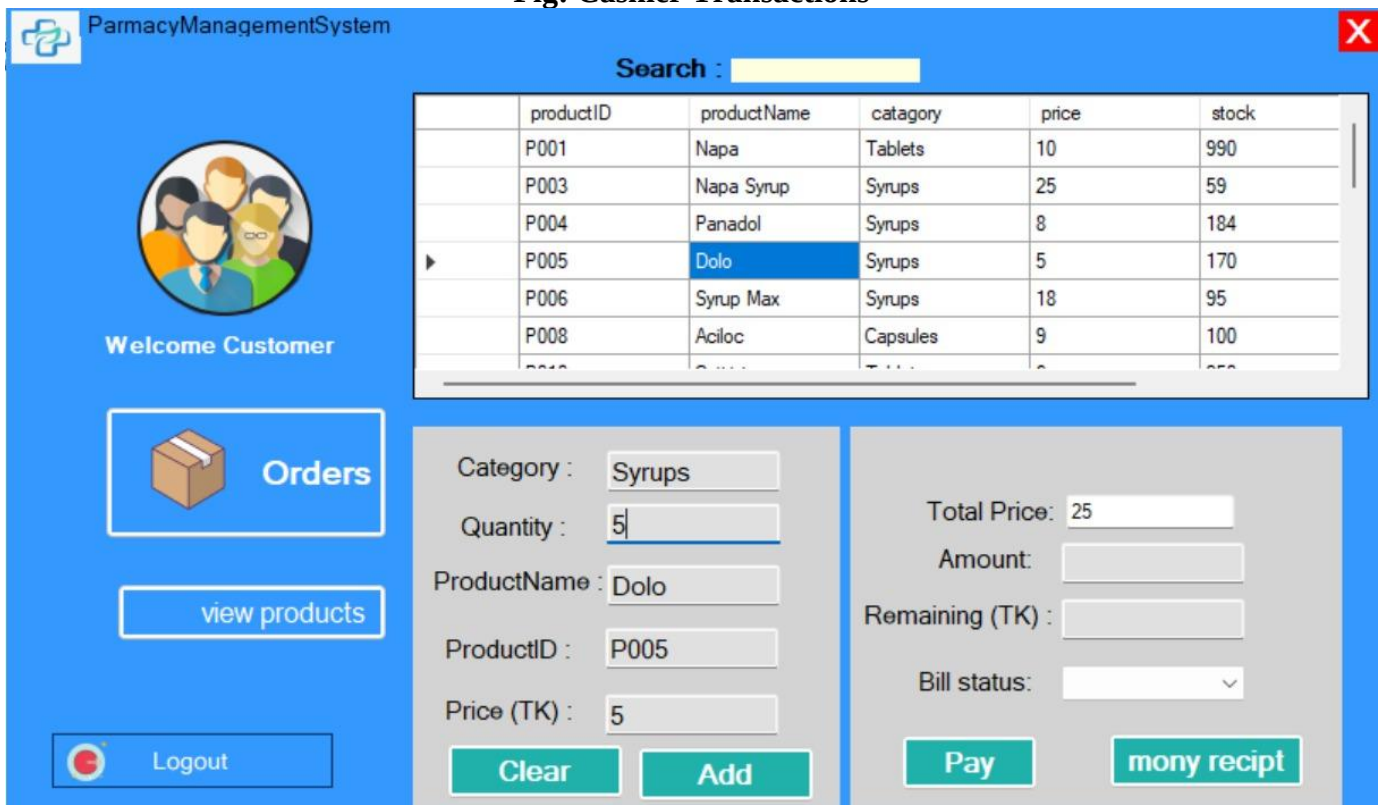


Fig: Customer View Product

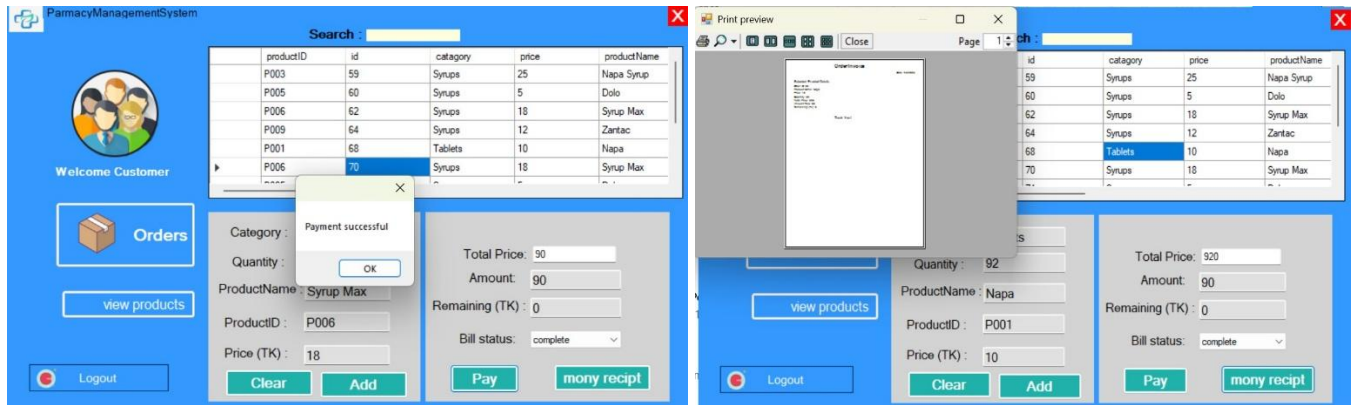


Fig: Customer Order

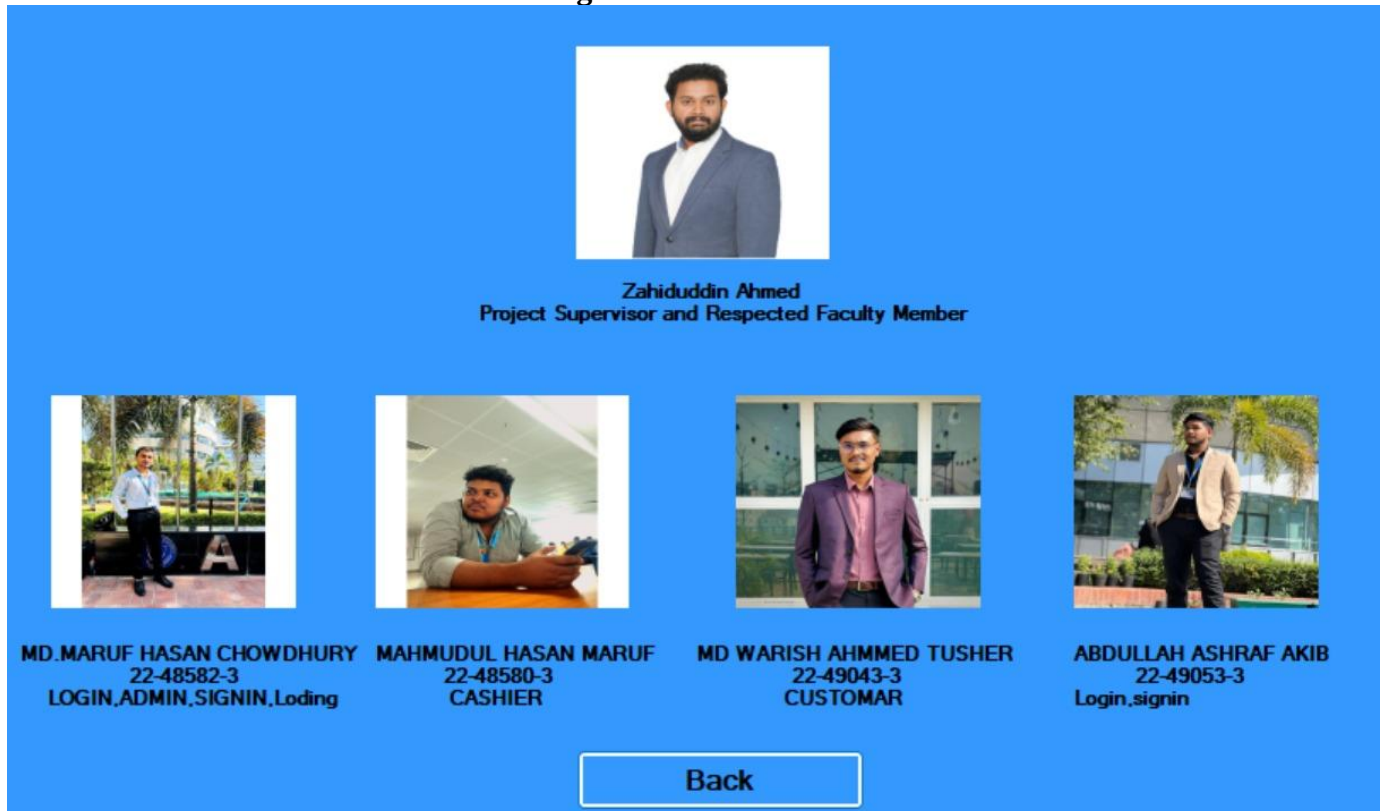


Fig: About us

